

University Rover Challenge 2027 – Requirements and Guidelines

The Mars Society's University Rover Challenge challenges students to build remotely operated rovers that can accomplish a variety of tasks that might one day assist astronauts working on the surface of Mars. Rovers will compete in four missions: 1) an Astrobiology Mission to investigate a site for the presence of life; 2) a Delivery Mission to deliver a variety of objects to astronauts in the field across rugged terrain; 3) an Equipment Servicing Mission to perform dexterous operations on a mock lander using a robotic arm; and 4) an Autonomy Mission to autonomously assist an astronaut and traverse a hilly area. A drone (aerial vehicle) may be used to assist the rover in the Delivery and Autonomy Missions, and will provide an advantage, but is not required.

The 2027 University Rover Challenge will be held June 2-5, 2027 at the Mars Society's Mars Desert Research Station (MDRS) near Hanksville, Utah, USA. The competition is open to both graduate and undergraduate students, and teams are permitted to include secondary (high school) students.

Any issues not covered by these published rule sets will be addressed on a case-by-case basis by the University Rover Challenge (URC) Director. Please consult the Questions and Answers (Q&A) portion of the URC web site (<http://urc.marssociety.org>) for updates. All matters addressed in the Q&A are applicable to the requirements and guidelines.

1. Competition Missions

1.a. The rover shall be judged in the four competition missions outlined below, and also on the System Acceptance Review (SAR – described in Section 3.a.iii).

1.a.i. Each event and the SAR shall be worth 100 points, for a total of 500 points. Penalties for overweight rovers, interventions, and other penalties are additive: e.g. penalties of 10% and 20% would result in a score of 70% of the points earned. Missions are scored independently and it is not possible to score less than zero on a mission.

1.a.ii. From the time teams are given access to their command and control (C2) station, they shall be able to set up all necessary systems, including all communications systems, and be **ready to compete in no more than 15 minutes**. Teams shall be able to **remove all equipment in no more than 10 minutes** at the end of the mission, and will be asked to switch off radio equipment immediately.

1.a.iii. **Teams do not need to return the rover or drone to the start gate, or collect any deployed items (e.g. radio repeaters, cameras, tools) before the end of time for any of the missions.** However, they must be returned/collected within the 10 minute limit after competing.

1.a.iv. For the four competition events, the rover is not required to be in the same configuration so modular pieces can be swapped between missions. On days that teams compete in the Astrobiology and Delivery Missions, teams will only compete in one Mission. Teams may be required to begin the Equipment Servicing Mission as soon as 10 minutes after the completion of the Autonomy Mission, operating from the same C2 station on an adjacent course.

1.a.v. The rover will be accessible throughout the competition and modifications can be made at any point.

1.b. Astrobiology Mission

The goal is to collect sample(s) chosen by the team at sites selected in the field, perform basic science evaluation of these samples with onboard instrumentation, and store at least one sample as a cache. Sites shall be analyzed for their likelihood to support microbial life with at least one life detection method. The team should evaluate at least two sites and acquire samples from one site to use as the returned cache.

1.b.i. Teams shall submit a written Science Plan by May 21, 2027, which will be factored into the judges' evaluation and score for the Astrobiology Mission. This will expand upon the science plan submitted in the System Acceptance Review (see Section 3.a.iii). Specifications for the plan will be posted to the URC website.

1.b.ii. Teams will be given a field briefing by the judges to discuss the tasks at the Astrobiology site. Through the information relayed by the rover, teams shall then select sites of potential biological interest within a 0.5 km radius of the C2 station. Teams will be given between 20 and 30 minutes to collect and analyze data with the rover, and may visit as many sites as time allows.

1.b.iii. Teams shall document each site investigated by:

- A wide-angle panorama showing the full context of the site. The panorama must indicate cardinal directions and have some indication of scale.
- A close-up, well focused, high-resolution picture with some indication of scale at the sampling sites.
- GNSS coordinates of each site, to include elevation and accuracy range.

Thorough documentation is especially crucial for the sample that is returned onboard the rover.

1.b.iv. The rover may use cameras or other sensors to investigate the area and may collect regolith/soil for analysis by instrumentation on-board the rover. Samples must be analyzed by the rover on-site and may not be brought back to the team for investigation or laboratory analysis.

1.b.v. Onboard equipment at a minimum should include:

- Two life detection capabilities. At least one life detection method shall be a bioassay.
- Additional sensors, subsystems, and test procedures are left to the discretion of the teams to meet the science-driven objectives of this task.

1.b.vi. Based upon investigation of at least two sites, teams shall then **collect a sub-surface sample from one of the sites at a depth of 10 cm or deeper. Sample(s) must be at least 5g** and may consist of a single rock, loose soil, or anything in between. Sample(s) may return the full depth including the topsoil but teams must be able to distinguish the soil depth for any sample. **The portion of the sample from below 10 cm will be used to determine the sample mass.**

1.b.vii. After collection of the sample, the rover must then store and close/seal it in a cache container onboard the rover, and return the sealed cache to the C2 station. The sealed cache

must prevent the sample from spilling if laid on its side or upside down. The cache must be removable from the rover within 5 minutes after the end of roving time. Team members must remove the cache and hand it to the judges. The cache must reasonably be able to be opened by the judges.

- 1.b.viii. Multiple samples may be collected (to perform different tests on) but only one sample is required to be cached. The cached sample does not need to be a pristine sub-surface sample, but may have had tests performed on it such as the addition of water or other chemicals.
- 1.b.ix. Any chemicals used onboard, including water and any reaction products, must follow a **no-spill policy** of being contained on the rover and not spilled on the ground. **Use of hazardous chemicals must be pre-approved prior to competition by submitting a plan of transportation, usage, disposal, safety precautions, and accident plan.** Teams should consider that URC takes place in a remote desert location with very limited water supplies and no quick access to emergency medical care.
- 1.b.x. After giving the cached sample to the judges, teams will have a debrief and discussion with the judges. **Discussion with the judges will occur even if the rover was unsuccessful in performing any of the field tasks.** The discussion should include:
 - Results of on-board rover tests performed.
 - Method used to ensure the sample was collected at least 10 cm below the surface and transferred to the judges without contamination.
 - Reasoning for sample site selection and documentation of each of the two sites.
 - Meaning of the data collected with respect to the presence or absence of life and implications for the habitability potential of the site.
 - Applicability to Mars.
- 1.b.xi. The score for this task will be based on the following components:
 - Thoroughness of the investigation of sites (panoramas, site selection).
 - Quality and applicability of the onboard analysis (life detection and science capabilities of choice).
 - Quality of the sample returned (weight, depth, possible contamination).
 - Scientific knowledge of astrobiology.

1.c. Delivery Mission

- 1.c.i. This will be a staged mission in which rovers shall be required to assist astronauts in the field, such as by finding, picking up, and delivering objects, all while traversing terrain of increasing difficulty. In this mission **the rover may be assisted by a drone, but not by any other mobile deployable systems (i.e., a mini-rover).** Teams will be given a fixed amount of time for each stage. Each stage will include multiple tasks as described below, and teams must achieve a specified minimum score within the allotted time on Stage 1 in order to proceed to Stage 2. Any time remaining at the completion of Stage 1 is added to the time allowed for Stage 2. Total on-course time will be between 30 and 60 minutes. A script describing the course stages and tasks to be accomplished will be published prior to the competition.

- 1.c.ii.** The natural terrain around MDRS includes soft sandy areas, gravel, rough stony areas, rock and boulder fields, vertical drops and steep loosely consolidated slopes. Terrain will range from flat close to the starting line, to exceedingly difficult obstacles at distances of up to 1 km from the start gate also involving navigation challenges. Portions of this mission, in later tasks, will be intentionally placed beyond direct line-of-sight of the C2 station antenna.
- 1.c.iii.** The rover teams can expect a combination of the following tasks:
- Open boxes with hinged lids such as toolboxes.
 - Pick up, carry, and deliver objects to a different location. Objects to be picked up and delivered will be in a variety of sizes, shapes, and weights. Objects may consist of small lightweight hand tools or instruments, rocks, or supply containers. Objects may be up to 5 kg in mass and will be smaller than 40 x 40 x 40 cm. All objects will have some sort of graspable feature, that is no greater than 7.5 cm diameter, varying in complexity from a handle to natural features.
 - Read signs carried by astronauts or placed in the field.
 - Find a piece of equipment in a large search area.
 - Find objects in an area with poor radio reception. Sought objects will be readily identifiable.
- 1.c.iv.** Approximate Global Navigation Satellite System (GNSS) coordinates will be provided for each pickup/delivery location. In certain cases, specific instructions will be provided for each object in advance, and in other cases, the object(s) to be delivered will be indicated at the delivery location (for example, on a small sign held by an astronaut). **Working GNSS is essential for this mission.**
- 1.c.v.** Teams will be scored on their ability to pick up and deliver the correct objects to the correct locations. Points may be awarded for partial completion of any particular task.

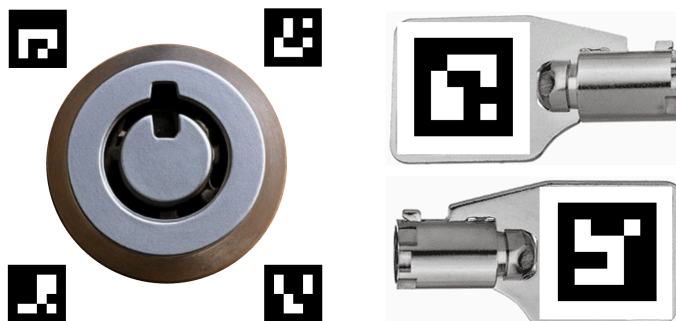
1.d. Equipment Servicing Mission

- 1.d.i.** Teams may be required to begin this mission as soon as 10 minutes after the completion of the Autonomy Mission, operating from the same C2 station on an adjacent course. Rovers shall be required to perform several dexterous operations on a mock-up lander. The rover shall have to travel up to 100 m across relatively flat terrain to reach the lander. Equipment requiring interaction will be between 1.5 m height and the ground. Teams will have 30 minutes to complete the mission and receive points for every task completed successfully.
- 1.d.ii.** The Equipment Servicing Mission will involve delivering a cached science sample to a lander and performing maintenance on the lander. It will include the following sub-tasks which may be performed in any order:
- Pick up a sample tube (test tube sized) and insert it into a larger cache container, close the lid, and turn the lock.
 - Pick up the cache container and transport it to the lander. The cache will have a handle at least 10 cm long and not more than 5 cm in diameter. The cache will weigh less than 5 kg.
 - Open a drawer on the lander. Insert the cache into a tight-fitting space in the drawer, and close the drawer.

- Undo a latch on a hinged panel of the lander and open the panel.
- Autonomous Typing: Team operators will be given a 3-6 letter launch key prior to the start of the mission which they may program into the rover prior to the start of this task. The rover must autonomously input this launch key on a keyboard. Typing must be correct for full points, but spelling mistakes or repeated letters may be corrected using the backspace/delete button or ignored for partial points. Teams must clearly state to the judge in the C2 station when entering or leaving autonomous mode, and be hands-off the controls during autonomous operations. Operator intervention is allowed only to exit autonomous mode and abort or restart the attempt on this task from the beginning. There is no penalty for aborting an attempt, and the rover may have multiple attempts at entering the launch key. The keyboard will be vertically mounted and specifically is [this keyboard](#) and [this display](#). 2 x 2 cm ArUco fiducial markers (from the 4x4_50 library) will be placed by the corners of the keyboard to help aid with alignment.



- Remove a key from a panel mounted tubular pin tumbler lock and insert into a different lock and turn the key. ArUco fiducial markers 1 x 1 cm will be placed on the key and by the corners of the keyhole, similar to the following:



- Connect [a hose](#) from a fuel tank to the lander. The connector will be a 1.5 inch diameter cam lock hose fitting with the internal sealing gasket cut down to half the original thickness. This is a push-on connector.
- Turn a ¼-turn lever valve handle.
- Push buttons, flip switches, turn knobs.

1.e. Autonomy Mission

- 1.e.i. Total time on course will be 40 minutes. The autonomy mission will consist of two sub-missions: i) assisting an astronaut in the field, ii) route finding to determine the easiest route to a location. Sub-missions may be attempted in any order. A drone may be used for reconnaissance while the rover is assisting the astronaut.
- 1.e.ii. There must be an LED indicator on the back of the rover, visible in bright daylight (e.g. LED array or high power LED), that will signal:
 - Red: Autonomous operation
 - Blue: Teleoperation (Manually driving)
 - Flashing Green: Successful arrival at a target.
- 1.e.iii. While stopped at any time, teams may do any programming including entering GNSS points, waypoints, or stay-out zones, and making changes to the controls and algorithms, but may not drive the rover.

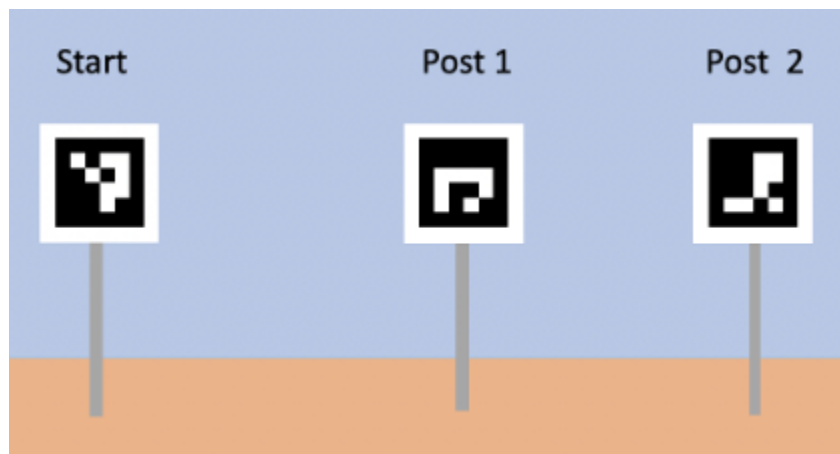
Astronaut Assistance Sub-mission (total 50 points).

- 1.e.iv. Teams will provide a team member who will act as an astronaut in the field. The astronaut must have an EVA-suit provided by the team. The suit does not need to be accurate for use on Mars. I.e. it does not need to be a pressurized or compression garment, and does not need to include a backpack and oxygen tanks. The suit must include a camera and a microphone that can be monitored by the operators in the command and control station (5 points). Please consider the hot desert environment when making your EVA suit. If a helmet is included it must be easy to remove by the astronaut if they become hot or uncomfortable.
- 1.e.v. Autonomously drive to a provided GNSS location where the astronaut is waiting. Stopping within 3 m of the GNSS location will be considered successful. (5 points)
- 1.e.vi. **Follow!** Astronaut will give a command to the rover to follow them and will then walk to a location given during set-up time. The rover must autonomously follow the astronaut, stopping within 3m of the astronaut when the astronaut stops. (15 points maximum):
 - Command given by a device carried by the astronaut (5 points (1/3 points))
 - Visual recognition of a sign carried by the astronaut. Sign may have any image chosen by the team including AR tags, written words, or pictures. (5 points (1/3 points))
 - Audio recognition of a word or phrase such as “follow” as said by the astronaut. (10 points (2/3 points))
 - Visual recognition of a beckoning gesture quietly given by the astronaut (15 points (full points))
- 1.e.vii. **Stay!** Astronaut will give a command to stay where it is while the astronaut moves to a new location >20m away. Rover should not move until given another command. (5 points)
- 1.e.viii. **Fetch!** Astronaut will give a command to the rover to fetch a tool. Autonomously pick up a rock pick hammer ([Amazon link](#)) (10 points). Hammer may be picked up using teleoperation for no points.
- 1.e.ix. **Come!** Astronaut will give a command for the rover to autonomously drive to the astronaut. Stopping within 3 m of the astronaut will be considered successful. (5 points).
- 1.e.x. **Give!** On command, autonomously place on the ground or drop the hammer. (5 points)

1.e.xi. At any point during the Astronaut Assistance Sub-Mission the rover may autonomously either stop or return to the astronaut for no penalty, and the command may be re-given. The operators in the C2 station may also send a signal to the rover to either stop or return to the astronaut for a 20% penalty on that individual task (each of 1.e.v through 1.e.x), or teleoperate back to any previously visited location for a 50% penalty on that individual task. The command may be re-given and the task re-attempted. Penalties for exiting autonomous mode will be capped at 50% so further abort commands or teleoperation is penalty free and will only consume time not points. Physical intervention penalties (see Section 3.e) are not capped.

Autonomous Route-Finding sub-mission (total 50 points).

1.e.xii. Teams will be given GNSS coordinates at a mission start location on flat easy terrain and may teleoperate to that location. Teams will be given two target locations in a hilly area that the rover must autonomously navigate to. Teams may attempt the targets in any order and do not need to return to the start location. Targets will be marked with AR posts as shown below. The posts will have 3-sided visual markers with 20 x 20 cm faces, 0.5 – 1.5m off the ground. **Each marker will display a black and white ArUco tag using the 4x4_50 tag library exactly as shown below.** Each tag will have a white border 1 cell in width, so cells are 2.5 cm across. The same tag will be displayed on each of the 3 sides so that it is visible from any direction.



- 1.e.xiii.** Locations will be in the Utah state-owned square mile between approximately (38.411N, -110.786E) and (38.425N, -110.768E). Teams are encouraged to use satellite imagery and maps including Digital Elevation Maps (DEM) of the area. URC will attempt to provide teams with a 1m or less resolution DEM of the area.
- 1.e.xiv.** A drone may be used for reconnaissance while the rover is assisting the astronaut, and may teleoperate the drone, but must land at the designated landing pad before the rover proceeds from the mission start location to the target locations.
- 1.e.xv.** One target location will be on a hill requiring teams to find an easy route up the hill. It will be intentionally selected so that not all approach directions will be navigable.

- 1.e.xvi.** The second target location will be intentionally placed behind the hill out of radio communications line of sight with the C2 station.
- 1.e.xvii.** 25 points per target location reached by the rover and autonomously stopping within 1m of the target location and indicating its arrival. Partial points will be given for successfully autonomously completing part of the route.
- 1.e.xviii.** At any point in the route-finding sub-mission operators may send a signal to the rover to stop and autonomously return the way the vehicle came by any distance or directly to the mapping start point at a penalty of 20% on the points for whichever target location is being attempted. Operators may also teleoperate back to any previously visited location for a 50% penalty on whichever target location is being attempted. Teleoperation may not go into any areas not already autonomously explored by the rover. Penalties for exiting autonomous mode will be capped at 50% so further abort commands or teleoperation is penalty free. Physical intervention penalties (see Section 3.e) are not capped.

2. Equipment Rules

2.a. Rover Rules

- 2.a.i.** The rover shall be a stand-alone, off-the-grid, mobile platform. Tethered power and communications are not allowed to the command and control (C2) station, but are allowed between the rover and any deployable items. A single rover carrying all deployable items, except the drone, must leave the designated start gate. In the open field the rover may deploy any number of smaller sub-platforms, except as restricted in 1.c.i for the Delivery Mission, so long as the combined system meets all additional requirements published.
- 2.a.ii.** Rovers shall be weighed and measured by the judges during the set-up time of each mission. For measurement the rover **must fit completely within a 1.2 m x 1.2 m x 1.2 m box**. Rovers may be placed in any orientation, and articulate/fold/bend to fit within the “transport crate,” but may not be disassembled to do so. This includes wheels, antenna, and any other system protruding from the rover. **Failure to fit within the specified dimensions at weigh-in will result in a 40% penalty. After weighing, rovers may unfold/expand to any size.**
- 2.a.iii.** The maximum allowable mass of the rover when deployed for any competition mission is **50 kg**. The total mass of all fielded rover parts for all events is **70 kg**. For example, a modular rover may have a robotic arm and a science module that are never on the rover at the same time. The combinations of rover plus arm and rover plus science module must each be under 50 kg, but the total rover plus arm plus science module must be less than 70 kg.
- The weight limits do not include any spares or tools used to prepare or maintain the rover, but does include any items deployed by the rover such as sub-rovers, cameras, communication relays.
 - Rover weight will be rounded down to the nearest whole kilogram.
 - For each event in which the rover is overweight, the team shall be assessed a **penalty of 5% of the points scored, per kilogram over 50.**

- 2.a.iv. Rovers shall utilize power and propulsion systems that are applicable to operations on Mars. Air-breathing systems are not permitted: No power or propulsion system may ingest ambient air for the purpose of combustion or other chemical reaction that yields energy.
- 2.a.v. **All rovers shall have a red push button emergency stop that is readily visible and accessible on the exterior of the rover.** It must be a push button, not a knob or rotating switch. This push button shall immediately stop the rover's movement and cease all power draw from any and all batteries in the event of an emergency such as a battery fire. All systems on the rover must have the power switched off by this single E-stop.
- 2.a.vi. Any mobile deployables such as a mini-rover require their own E-stop, with the exception of the drone. A fuse will suffice for any devices that cannot move on their own.

2.b. Drones

- 2.b.i. **Use of a drone will be advantageous for the Delivery and Autonomy Missions, but full points will be possible without a drone.** A drone may be especially useful in reading signs, searching large areas, picking up and delivering objects, acting as a radio relay, or operating in areas where a rover on the ground may have poor radio reception. Not all objects will be of a size or weight appropriate to be picked up and carried by <5kg drones.
- 2.b.ii. Drones will be remotely operated by a pilot in the command and control (C2) station, and will need radio communication with the drone via an antenna outside of the C2 station, the same as the rover.
- 2.b.iii. Drones must be of a rotary wing design and able to hover in a fixed location. No lighter-than-air vehicles are allowed as they are impractical on Mars, and fixed-wing drones are being excluded for safety reasons.
- 2.b.iv. Drones will start at a designated landing pad in the field. After initial launch drones may take off and land as many times as desired following rules given by the judges.
- 2.b.v. To simulate flight on Mars the **drone must carry a dummy mass of the same weight as the battery. It may NOT be a spare battery.** The dummy mass must be inert and of no functional use. **The dummy mass and battery must be easy to remove for weighing.** The dummy mass must be carried at all times regardless of wind conditions
- 2.b.vi. It is strongly suggested that drones should be able to operate in a 48 kph / 30 mph wind. This requires a top speed of least 64 kph / 40 mph for safe operation. Teams may fly their drone in any wind speeds they feel confident it can handle. Teams should expect highly variable wind conditions throughout URC and are not guaranteed that they will be scheduled during a period of low wind. In the event of high winds the dummy mass requirement will NOT be waived.
- 2.b.vii. Drones are not included in the weight and size restrictions for the rover. **Drones may not exceed 5 kg (11 lb) including the dummy mass. Drones exceeding the 5kg weight limit will not be permitted to be used in the URC competition** for safety reasons. **All drones in excess of 250 g (0.55 lb) must be registered with the United States Federal Aviation Administration (FAA) DroneZone and marked with the registration number.**
- 2.b.viii. The drone's cost must be accounted within the budget limit as defined in Section 3.c.
- 2.b.ix. **Drones must be equipped with a Remote ID meeting FAA requirements.**

- 2.b.x. All pilots must complete the FAA TRUST (The Recreational UAS Safety Test), and carry proof of passing with them.** It is a free online training course that takes about 30 minutes, and is valid for 3 years.
- 2.b.xi.** Airborne vehicles must meet any and all FAA guidelines that apply to recreational drone operation. It will be the responsibility of each team to research and comply with FAA requirements. The area around MDRS is class G (uncontrolled) airspace, and drones are required to **fly below 400 feet (120 m) above ground level**. Please beware that MDRS is fairly close to Capitol Reef National Park, and **drones are not permitted in National Parks**. It is also about 5 miles from the uncontrolled Hanksville airport, and drones must give way to any manned aircraft.
- 2.b.xii. A spotter must maintain visual line of sight at all times.** The pilot will not have a direct view of the drone so a designated team member in the field must act as a spotter. As well as an FAA requirement, the role of the spotter is to ensure the drone does not endanger anyone on the course. The spotter will use radio communication with the pilot to avoid unsafe behavior of the drone, or may take direct control of the drone if a dual-pilot setup is used. The spotter may only intervene in the event of potentially unsafe operation, and may not otherwise participate in the mission. Judges may require the spotter to intervene if safety is felt to be compromised.

2.c. Communications Equipment

- 2.c.i.** The rover (and drone) shall be operated remotely using wireless communications with no time delay. The operators will not be able to directly view the rover or the site, and line-of-sight communications are not guaranteed for all of the missions. Normal wireless internet is not available in the field or at MDRS, and use of satellite internet is not allowed. Teams are required to power down communications equipment at the event sites while not competing, so as not to interfere with other teams.
- 2.c.ii.** Wireless communication methods used by teams shall adhere to all applicable FCC (United States Federal Communications Commission) standards and regulations. **Teams must submit details regarding communication devices and operator licenses (when applicable) to the URC Director no later than Friday, April 30, 2027.** Team members are permitted to obtain and utilize any relevant licenses, but must document the license, applicable regulations, and devices as part of the communications documentation deadline. Teams must notify the URC Director immediately of any changes after this date.
- 2.c.iii.** Both omnidirectional and directional antennae are allowed, but communications equipment must not rely on the team's ability to watch and track the rover firsthand. Steered directional antennae may use a mechanized antenna mounted outside that is controlled via an electronic signal from the C2 station. Signal strength, relayed GNSS, or other strategies may be used to give feedback on antenna direction, but it is not allowed to mount a camera on top of the antenna for visual feedback.
- 2.c.iv. Base station antenna height is limited to 3 m** and shall adhere to all applicable regulations. Antenna bases must be located within 5 meters of the team's C2 station, and any ropes or wires used for stability purposes only may be anchored within 10 meters of the C2 station.

The exception to this is the use of structures at MDRS where allowable antennae locations will be given by the judge and may be located up to 20 m away from C2 stations to avoid underground pipe and cables, and other structures which may block radio signals. **All teams should bring at least 25 m of communications cable.**

- 2.c.v. Restrictions on the 900 MHz band (902-928 MHz):** Teams shall not use frequency bandwidths greater than 8 MHz. **Teams must also be able to operate exclusively within each of the following three sub-bands: "900-Low" (902-910 MHz), "900-Mid" (911-919 MHz), and "900-High" (920-928 MHz). The competition schedule will notify teams which sub-band may be used for each mission, and teams must be able to shift to another sub-band as required.** There is no limit on the number or width of 900 MHz channels a team uses, so long as they are all within the designated sub-band. **These restrictions apply to all communications between any of the C2 station, rover, drone, and astronaut, including any local area network (LAN) between devices at any location/vehicle.** There will be spectrum monitoring on-site to ensure that teams are not interfering with channels outside those allotted. Teams should anticipate being within signal range of other teams operating on different 900 MHz sub-bands.
- 2.c.vi. No restrictions on the 2.4 GHz band.** Frequency restrictions in the 2.4 GHz band have been removed. Within the 2.4 GHz band teams may use the band they chose, providing they comply with FCC requirements. Teams must expect to be operating within close proximity of other teams and be able to deal with any potential interference. Teams are strongly encouraged to investigate frequency hopping, automatic channel selection, automatic channel switching, and other interference-tolerant protocols. Teams are also highly encouraged to operate in bands outside of 900 MHz and 2.4 GHz and to obtain ham radio licenses to allow higher power operation on less used bands.

3. Competition Rules

3.a. Schedule

Prospective teams will undergo a review and down-selection process, meaning that only teams who pass each milestone will be invited to compete in the field. Teams failing to qualify for the field competition are strongly encouraged to enter other Rover Challenge Series events. Specific details for each deadline (including deliverable format, submission requirements, and judges' expectations) will be posted to the URC web site (<http://urc.marsociety.org>). Judges may respond to teams with follow-up questions or requests for clarification at any of these milestones.

3.a.i. Registration.

Teams are required to register and declare their intent to compete **no later than Friday, October 30, 2026.** No significant deliverables are required for this deadline, aside from team details requested via the URC web site.

3.a.ii. Preliminary Design Review

Teams are required to submit a Preliminary Design Review (PDR) document no later than Wednesday, December 2, 2026. The PDR document must include the team structure, resources, and project management plan (including a Gantt chart, initial budget, fund-raising

plans, recruiting, and educational outreach). Technical details regarding the rover should include the current state of design, development, and prototyping. Judges will be assessing each team's overall likelihood of being ready in time to compete in the URC competition. Teams will be assessed on their own merits, not against other teams.

3.a.iii. System Acceptance Review

Teams are required to submit a System Acceptance Review (SAR) no later than Friday, March 5, 2027. The SAR will focus on demonstration of the capabilities and customization of the rover systems/sub-systems to perform each of the missions. This includes the overall system design, and progress to-date of the final system. The SAR will consist of both written and video components. The SAR is a competitive milestone and packages will be judged against other teams' submissions by the judges. The 36 teams who score the highest in the SAR milestone will be invited to compete in the field.

3.a.iv. Field Competition

June 2-5, 2027 at the Mars Society's Mars Desert Research Station (MDRS) near Hanksville, Utah, USA.

3.b. Team Members

- 3.b.i.** There is no restriction on the number of team members allowed during the year. Students must be enrolled at least half-time in a degree or high school diploma granting course. Students from multiple universities may compete on the same team. A single university may field multiple rovers and multiple teams, however there may be no overlap between team members, budget, donated equipment, or purchased equipment.
- 3.b.ii.** Teams are encouraged to work with advisors. Advisors should limit their involvement to academic level advising only. Nontechnical management duties, including tracking finances, registration, fundraising, websites, promotion, submission of deliverables, and communication with URC staff, fall within the duties of the students. Advisors can spectate from the field, but may not spectate from within the C2 station.
- 3.b.iii.** Team attendance limits for the URC Finals will be enforced due to the permitting, logistical, and environmental impact considerations associated with managing a large field event in a remote location. Detailed attendance limits are found on the URC website.
- 3.b.iv.** All team members operating the rover must remain in the designated operators' area (C2 station). Nobody may follow alongside the rover for the purpose of providing feedback to the operators. Members of the judging team, media, non-operator team members, and other spectators may only follow a rover at the judges' discretion. Team members following the rover may participate as runners in accordance with Section 3.e, or activate an emergency kill switch (in the event of an emergency), but may not otherwise participate in that mission.
- 3.b.v.** It is incumbent upon the student team leaders to ensure that their respective teams uphold the integrity of this competition.

3.c. Finances

- 3.c.i. Teams shall be required to track all finances as related to this project, documented in the Project Budget Table, which is included in the PDR and SAR submissions.
- 3.c.ii. In addition, each team must **submit a final Rover Expense Report no later than May 26, 2027** (if necessary, teams may submit an updated record on the first day of the URC event – June 2, 2027). Teams shall be penalized 10% of total points per day if they are late in submitting the expense report, and will be disqualified for not submitting their expense report by the end of the URC event (June 5).
- 3.c.iii. The maximum allowable cash budget to be spent on the fielded equipment is **\$26,000 US**, which shall include any fielded components for the rover, rover modules, rover power sources, rover communications equipment, drone, astronaut EVA suit, and base station equipment including the antennae, transceivers, and all C2 equipment such as computers, monitors, controllers, etc.
- 3.c.iv. The Director may allow certain sponsorships that are available to all teams to count as an extension of the rover budget limit.
- 3.c.v. The budget limit pertains only to fielded equipment and does not include other project expenses such as travel, accommodation, equipment transportation, tools, prototype parts not used in the field competition, and spare parts that can be replaced one-for-one in the case of damage to the original.
- 3.c.vi. If used equipment is purchased commercially the as-bought price may be used. Re-used equipment from prior competitions must be valued at either the original as-bought cost, or the current cost for a new version of the same or equivalent item.
- 3.c.vii. Any rented equipment must be valued at purchase cost (new or used).
- 3.c.viii. Shipping and taxes must be included in the cost since these are a standard part of the cost of any item.
- 3.c.ix. Corporate sponsorship is encouraged. If equipment or services are donated to the team either free or at reduced cost, the full cost of a new or second-hand component must be used. Donations must be documented by the donor, but teams may use the cheapest rate commercially available for the same equipment or service.
- 3.c.x. Non-US teams have an allowable rover budget equivalent to \$26,000 US based on the most advantageous documented currency conversion rate between August 1, 2026 and June 2, 2027.
- 3.c.xi. Teams may be required to submit receipts as proof of the rover expenses upon request.

3.d. Operations

- 3.d.i. Teams will operate their rovers in real-time from designated command and control (C2) stations. These stations will be metal trailer units (such as the back of a small moving truck provided by URC) or structures at the Mars Desert Research Station. **Visibility of the course to the operators in the C2 station will be blocked.** Basic power (120V, 60Hz), tables, and chairs will be provided. All of the competition events will be held in full daylight.
- 3.d.ii. There should be radio communication line-of-sight from the C2 station to the rover for the Astrobiology and Equipment Servicing Missions. For the Autonomy mission there will likely

not be line-of-sight communications for all areas in the mapping sub-mission. For the Delivery Mission, line of sight communication is not guaranteed for more than 50% of the course, and will be intentionally out of line of sight for later tasks. Rovers are not expected to travel as far as 1 km from the C2 station.

- 3.d.iii.** Temperatures at MDRS can reach 100°F (37.8°C) and winds frequently whip up dust. Rovers shall be able to withstand these conditions and also light rain, but will not be expected to compete in heavy rain or thunderstorms.
- 3.d.iv. Testing is not allowed at MDRS before, during, or after URC.** Teams may test in town or at other nearby sites where off-road vehicles are allowed (e.g. [Factory Butte OHV Open Area](#)). Land controlled by the Bureau of Land Management, that is not specifically designated for off-road use, is strictly not allowed for any URC purposes.
- 3.d.v.** The Global Navigation Satellite System (GNSS) standard shall be the WGS 84 datum. Coordinates will be provided in latitude/longitude format (e.g. decimal degrees; degrees decimal minutes; degrees minutes seconds).

3.e. Interventions

If a rover, drone, or other deployable suffers a critical problem during a mission that requires direct team intervention (anything requiring touching of the devices in the field), that intervention shall be subject to the following:

- 3.e.i. A request for an intervention can only come from the team members operating the devices, not any team members spectating in the field.** They may designate any number of team members who may go to repair or retrieve the device (hereafter referred to as “runners”). Spectating team members may be asked to act as runners, and also rover operators may leave the C2 station and become runners. Spectating team members may carry tools.
- 3.e.ii. If a spectating team member intervenes without request from the operators, it counts as an emergency stop.** This is allowed such as to rescue the rover to prevent a fall or a fire. The current mission will be considered terminated although the rover may compete in other subsequent missions. All points earned in a mission to this point are preserved, and in the Astrobiology Mission teams may still conduct their field briefing.
- 3.e.iii. If a team member leaves the C2 station to become a runner they will NOT be permitted to return to the C2 station** to participate in operating the mission, or analysis of any data, after this point for the current mission. Runners will still be permitted to retrieve or repair devices in future interventions.
- 3.e.iv.** Runners may fix the device in the field without moving it, or return it to the C2 station, or return it to the start of that obstacle/task as defined by the judge in the field. Judges may require it be moved for the safety of the team members or preservation of the course.
- 3.e.v.** If the device is returned to the C2 station, the operators may take part in the diagnostic and repair process, but runners and spectators may not communicate any details about the mission site to the operators.
- 3.e.vi.** During an intervention the team members in the field may communicate directly with the operators in the C2 station to facilitate repairs. **If teams wish to use radios for this purpose**

it is strongly recommended that they bring their own radios, but they may be used only during an intervention. Otherwise teams may borrow a judge's radio. **All radio communication must be in English** so judges can properly monitor conversations.

3.e.vii. If a drone or secondary rover is used and an intervention is required on any vehicle, teams may continue to operate the working vehicles. If the drone returns to the C2 station under its own power, it should land at the designated landing pad and be carried to the C2 station by a runner. Swapping batteries on either vehicle counts as an intervention.

3.e.viii. Teams will be **penalized 20% of the total points in that mission for every intervention.** The mission clock will continue to run during an intervention. Multiple intervention penalties in a single mission are additive: e.g. two interventions would result in a score of 60% of points earned.